

CPSC 250 Development Environment Setup

Programming for Data Manipulation

Overview

This document will help you set up the software development environment that we will use in CPSC 250. The goal is to make sure that you can:

- run Python programs on your own laptop,
- use PyCharm as your programming environment,
- access course materials from GitHub,
- use git to keep your copy of the course repository up to date, and
- install the Python packages used throughout the course.

This setup is important. If your environment is working correctly, the programming work in the course will be much smoother.

Important Concepts

Before starting, it is useful to distinguish among the main tools.

Tool	Purpose
Python	The programming language that executes your programs.
PyCharm	The development environment used to write, run, and manage Python programs.
Git	The version control system used to track changes in code.
GitHub	A website that stores git repositories online.

PyCharm helps you write and manage code, but Python is what actually runs the code.

1 Install Python

If you already have Python 3.9 or newer installed, you may skip this section. Otherwise, install Python using the instructions below.

Windows Instructions

1. Go to <https://www.python.org/downloads/>.
2. Download the latest stable Python 3 release for Windows.
3. Run the installer.
4. **Important:** before clicking Install, make sure you check the box labeled:

```
Add Python to PATH
```

5. Complete the installation.

Verify Python on Windows

Open **Command Prompt** and run:

```
python --version
```

You should see output similar to:

```
Python 3.12.2
```

The exact version number may be different. Any Python version 3.9 or newer is fine.

macOS Instructions

1. Go to <https://www.python.org/downloads/>.
2. Download the latest stable Python 3 release for macOS.
3. Run the installer.
4. Complete the installation.

Verify Python on macOS

Open **Terminal** and run:

```
python3 --version
```

You should see output similar to:

```
Python 3.12.2
```

The exact version number may be different. Any Python version 3.9 or newer is fine.

2 Install Git

Git is the version control system that allows you to download repositories, track changes, and synchronize your work with GitHub.

Windows Instructions

1. Go to <https://git-scm.com/download/win>.
2. Download Git for Windows.
3. Run the installer.
4. Follow the default installation choices unless instructed otherwise.

After installation, open **Command Prompt** and run:

```
git --version
```

macOS Instructions

On macOS, git may already be available. Open **Terminal** and run:

```
git --version
```

If git is not installed, macOS may prompt you to install the Apple command line developer tools. Follow the prompt. You may also install Git from <https://git-scm.com/download/mac>.

3 Install PyCharm Community Edition

1. Go to <https://www.jetbrains.com/pycharm/download/>.
2. Download **PyCharm Community Edition** for your operating system.
3. Install PyCharm using the default options.
4. Launch PyCharm.

PyCharm Community Edition is sufficient for this course. You do not need PyCharm Professional.

4 Create a GitHub Account

If you do not already have a GitHub account:

1. Go to <https://github.com>.
2. Click **Sign up**.
3. Create a free account using your CNU email address.
4. Verify your email address if required.

Please choose a professional and identifiable username if possible.

5 Access the CPSC 250 Course Repository

The CPSC 250 course repository is located at:

```
https://github.com/brash99/cpsc250
```

This repository contains course materials such as lecture notes, example code, and other files used in the regular CPSC 250 course.

Important Distinction

Repository	Purpose
cpsc250	Course notes, examples, and reference materials.
cpsc250L	Lab exercises and student lab work.

For the regular CPSC 250 course repository, you may clone the instructor repository directly. If your instructor asks you to maintain your own copy, you may instead fork the repository first and clone your fork.

For CPSC 250L lab work, you should fork the lab repository and clone your own fork.

6 Clone the CPSC 250 Repository Using PyCharm

Step 1: Open PyCharm

Launch PyCharm. From the welcome screen, choose:

```
Get from VCS
```

If a project is already open, choose:

```
Git -> Clone...
```

Step 2: Log Into GitHub Through PyCharm

PyCharm may ask you to authenticate with GitHub.

1. Click **Log In via GitHub** or the equivalent login option.
2. Complete the GitHub login process in your browser.
3. Authorize PyCharm or JetBrains if requested.

Once this is done, PyCharm can communicate with GitHub directly.

Do not create or paste a personal access token into the repository URL. PyCharm's GitHub login is the preferred method for this course.

Step 3: Clone the Repository

To clone the instructor's CPSC 250 repository directly, enter:

```
https://github.com/brash99/cpsc250
```

Choose a local directory where the repository should be stored. PyCharm may suggest a directory such as `PyCharmProjects`. That is fine.

Click **Clone**.

If PyCharm asks whether you trust the project, choose **Trust Project**.

7 Configure the Python Interpreter in PyCharm

PyCharm needs to know which Python installation to use for this project.

Windows

In PyCharm, go to:

```
File -> Settings -> Project -> Python Interpreter
```

macOS

In PyCharm, go to:

```
PyCharm -> Settings -> Project -> Python Interpreter
```

A Python interpreter may already appear. If not:

1. Click **Add Interpreter**.

2. Choose a local Python installation.
3. Select Python 3.9 or newer.

8 Use the Terminal

Professional programmers frequently use the terminal. PyCharm includes a built-in terminal window.

In PyCharm, open the terminal using:

```
View -> Tool Windows -> Terminal
```

Windows Commands

Try the following commands:

```
dir
python --version
pip list
git --version
```

macOS Commands

Try the following commands:

```
ls
python3 --version
pip3 list
git --version
```

9 Install Useful Python Packages

We will use several Python packages throughout the course, including:

- numpy
- pandas
- matplotlib
- scipy

You may install packages through PyCharm's package manager or through the terminal.

Terminal Installation on Windows

In the PyCharm terminal, run:

```
pip install numpy pandas matplotlib scipy
```

Terminal Installation on macOS

In the PyCharm terminal, run:

```
pip3 install numpy pandas matplotlib scipy
```

This process may take several minutes.

10 Create and Run a Test Program

In PyCharm, create a new file called:

```
helloworld.py
```

Add the following code:

```
print("Hello CPSC 250!")  
print("My Python environment is working.")
```

Run the file in PyCharm. You should see:

```
Hello CPSC 250!  
My Python environment is working.
```

11 Basic Git Workflow in PyCharm

The basic git workflow is:

```
Edit files → Commit changes → Push changes
```

Even when PyCharm provides buttons and menus, the underlying tool is git.

Commit Changes

After editing a file, choose:

```
Git -> Commit...
```

Select the files you want to include, write a meaningful commit message, and click **Commit**.

Example commit message:

```
Add hello world test program
```

Push Changes

To send committed changes to GitHub, choose:

```
Git -> Push...
```

12 Keeping the Repository Up to Date

Your instructor may add new examples, notes, or other materials to the CPSC 250 repository during the course.

If you cloned the instructor repository directly, update your local copy using:

```
Git -> Pull
```

If you forked the repository, update your fork using:

```
Git -> GitHub -> Sync Fork
```

The first time you do this, PyCharm may ask you to authorize JetBrains to access your GitHub account. Follow the prompts.

13 Important Warnings

- Do not paste personal access tokens into repository URLs.
- Do not upload ZIP files manually to GitHub.
- Do not drag and drop random files into GitHub as a substitute for git.
- Use git commit and git push to save and synchronize work.
- Keep your GitHub username and password private.

14 Troubleshooting

Problem: python command not found

On Windows, make sure Python was added to PATH during installation. You may need to reinstall Python and check the **Add Python to PATH** box.

On macOS, try:

```
python3 --version
```

Problem: pip command not found

Try:

```
python -m pip install numpy
```

On macOS, try:

```
python3 -m pip install numpy
```

Problem: PyCharm says no interpreter is configured

Go to:

```
Settings -> Project -> Python Interpreter
```

and select your installed Python version.

Problem: PyCharm cannot log into GitHub

Make sure you can log into <https://github.com> in a browser. Then return to PyCharm and try the GitHub login again.

Problem: Program will not run

Check that:

- the Python interpreter is configured,
- the file is saved,
- there are no typing mistakes in the code, and
- required packages are installed.

Final Checklist

Before you are finished, confirm that you can do the following:

- Open PyCharm.
- Open the CPSC 250 project.
- Run a Python file.
- Open the PyCharm terminal.
- Check your Python version.
- Check your git version.
- Install Python packages.
- Pull updates from GitHub.

Once these steps work, your CPSC 250 development environment is ready.